Hashir Aziz

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Education

National University of Sciences and Technology (NUST) Islamabad, Pakistan
Bachelor of Science in Computer Science | CGPA: 3.63 *2023 – 2027*
Relevant Coursework: Deep Learning, Parallel & Distributed Computing, Machine Learning, Artificial Intelligence

Experience

AI Engineering Intern Islamabad, Pakistan
Dataquartz *July – August 2026*

- Developed and deployed an end-to-end production voice AI agent on **LiveKit** (STT → LLM → TTS pipeline).
- Designed and built an Urdu TTS evaluation framework benchmarking 4 models (**Piper**, **Indic Parler-TTS**, **Higgs TTS 3**, **OmniVoice**) on the UrduSpeech dataset, scoring WER, CER and real-time factor.
- Served Higgs TTS 3 (4B) with **vLLM-Omni** on a shared mixed-GPU server (Tesla P100 + RTX 3090 Ti), and exposed it through an OpenAI-style speech endpoint.

Projects

DawaCheck: Drug Interaction Checker for Pakistani Medications | LangGraph, LangSmith, FastAPI, RabbitMQ

- Built a drug-interaction checker for Pakistani brand and generic names (**DRAP** → **RxNorm** → **openFDA**), with a two-stage **LangGraph** pipeline that verifies every LLM-cited sentence against FDA label text.
- Implemented async job processing (**FastAPI**, **RabbitMQ**, **Postgres** LISTEN/NOTIFY, **WebSockets**) for long-running interaction checks.
- Built a **LangSmith** eval suite over 422 drug pairs with three custom evaluators (interaction-detection accuracy, citation groundedness, severity-overstatement), reaching 76%, 82%, and 98% pass rates.

JobFlow (Autonomous Interview Research Agent) | LangChain, Groq, FastAPI, PostgreSQL, Next.js

- Wrote a manual ReAct tool-calling loop instead of using LangChain's **AgentExecutor**, enforcing a minimum-tool-usage stopping condition and injecting corrective feedback when the model attempted to finalize early.
- Built resilient scraping (requests/BeautifulSoup with a **Playwright** fallback for JS-rendered postings) and a validation layer that catches malformed or hallucinated tool calls.
- Streamed agent execution over **FastAPI SSE** and persisted full run traces to **PostgreSQL** for observability and debugging.

Hybrid GraphRAG Platform | FastAPI, Neo4j, Qdrant, Celery, Redis, YOLOX, Tesseract

- Built a dual-path RAG backend using **asyncio.gather** to run multi-hop **Neo4j** graph traversal and **Qdrant** semantic search concurrently, cutting retrieval latency to sub-15ms.
- Added a **Celery/Redis** task queue to offload document ingestion and prevent event loop blocking under load.
- Built an intent-based LLM routing classifier with **Pydantic** and Instructor to map queries to database schemas and reduce hallucination.

K-forge (Automated MLOps Platform) | FastAPI, Next.js, Docker, Kubernetes, ONNX

- Built an MLOps platform that turns an uploaded model binary into a deployed inference service: a **FastAPI** code-generation engine compiles it into a Python microservice and synthesizes its Dockerfile.
- Deployed the generated containers to **Kubernetes** to serve and scale model inference, taking a model from upload to a running endpoint without manual deployment steps.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C/C++
LLM & Voice AI: LangChain, LangGraph, LangSmith, Retrieval-Augmented Generation (RAG), Tool Calling, LLM Evaluation, Pydantic, Instructor, LiveKit, Speech-to-Text (STT), Text-to-Speech (TTS)
ML & Deep Learning: PyTorch, Transformers, Scikit-learn, ONNX
Data & Storage: PostgreSQL, Qdrant, Neo4j, Redis, FAISS
Backend & Deployment: FastAPI, Celery, RabbitMQ, WebSockets, Docker, Kubernetes, vLLM, Git